

Cardiac Segmentation Protocol from CT Data in 3D Slicer

Introduction

3D cardiac modeling based on clinical imaging requires accurate, anatomically precise, and reproducible segmentation of cardiac structures. This protocol presents a standardized, step-by-step workflow for segmenting cardiac chambers, with a specific focus on the left atrium, from computed tomography (CT) data using the open-source 3D Slicer platform.

The document details a complete medical image processing pipeline, encompassing raw data preparation and spatial anisotropy correction, automated segmentation using the "Grow from Seeds" algorithm, interactive boundary refinement, and morphological separation of the endocardial, epicardial, and myocardial wall surfaces. The resulting geometric models are exported in .vtk format and are fully optimized for downstream applications in computational electrophysiology, transmural fiber architecture mapping, and patient-specific atrial fibrillation analysis.

Designed with scientific reproducibility in mind, this protocol is intended for bioengineers, cardiology researchers, and clinical specialists involved in the development of patient-specific digital heart models.

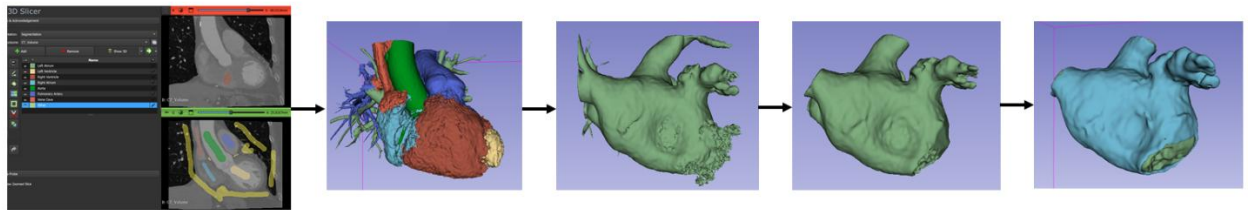


Figure 1. Segmentation pipeline illustration

Required Software and Data

- 3D Slicer: Download from <https://download.slicer.org/>
- Example dataset:
CT_Volume.nrrd (contrast-enhanced cardiac CT in Nearly Raw Raster Data format);
Segmentation_example.seg.nrrd (performed segmentation of a given CT volume)

Step-by-Step Protocol

1. Data Preparation

- Load the cardiac CT dataset into 3D Slicer. The preferred input format is .nrrd (Nearly Raw Raster Data), which preserves metadata and supports efficient volumetric processing.

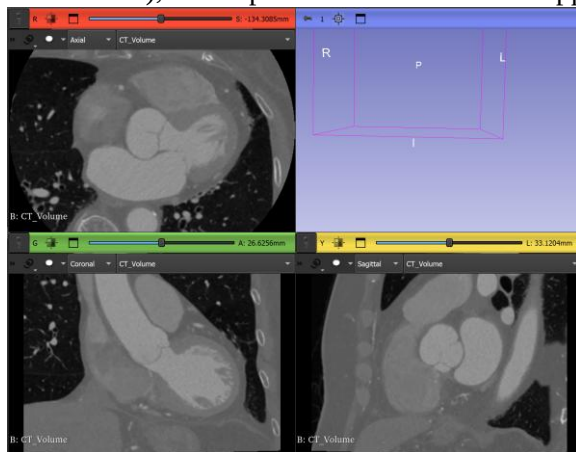


Figure 2. Visualization of a CT scan in the 3D Slicer program.

- Verify the voxel spacing (resolution) of the loaded volume. To do this:
 1. Navigate to the **Crop Volume** module.
 2. Create a **Region of Interest (ROI)** encompassing the entire cardiac anatomy.
 3. Inspect the **Volume Information** panel to examine the current spacing parameters.
Example: spacing = $0.4 \times 0.4 \times 0.625$ mm (indicating anisotropic voxel dimensions).
- If significant anisotropy is detected (i.e., voxel dimensions differ substantially across axes), apply Interpolated Cropping to resample the volume to isotropic resolution. Set the output spacing to a value intermediate between the minimum and maximum input spacing values.
Example transformation: $0.4 \times 0.4 \times 0.625$ mm $\rightarrow$ $0.4 \times 0.4 \times 0.4$ mm (reduced anisotropy)
- Apply the cropping transformation and hide the ROI to proceed with segmentation.

2. Initialization of Segmentation Structures

- Navigate to the Segment Editor module.
- Initialize a new segmentation node and create individual segments corresponding to the following anatomical structures:
 1. Left Ventricle (LV)
 2. Right Ventricle (RV)
 3. Left Atrium (LA)
 4. Right Atrium (RA)
 5. Aorta
 6. Pulmonary Artery
 7. Vena Cava (superior and inferior)
 8. **Auxiliary "Other" segment:** Assign a distinctive color for visual differentiation. This segment serves as a catch-all category for myocardial tissue, connective structures, and vascular elements not explicitly delineated in the primary segments

3. Seed-Based Annotation of Anatomical Regions.

Manual Seed Placement for Automated Segmentation

Utilize the Paint tool within the Segment Editor to place seed points (markers) within the interior regions of each target anatomical structure. These seeds serve as initialization points for the subsequent region-growing algorithm.

Recommended annotation sequence:

1. Left Ventricle
2. Aorta
3. Pulmonary Artery
4. Right Ventricle
5. Left Atrium
6. Right Atrium
7. Vena Cava
8. Other: Mark external/background regions and non-target tissues

Note: Precise boundary delineation at the seed-placement stage is not required, as the automated algorithm incorporates spatial regularization and smoothing. The primary objective is to ensure that each anatomical compartment contains at least one representative seed point.

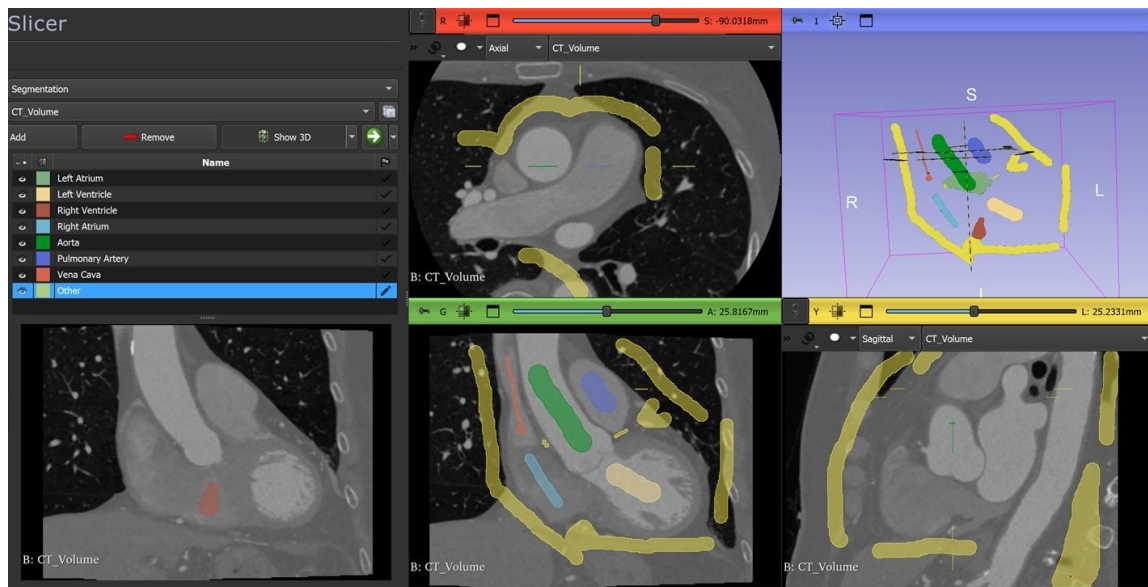


Figure 3. Setting markers of various anatomical structures of the heart for automatic segmentation. The colors indicate the corresponding regions

4. Automated Segmentation via "Grow from Seeds"

Algorithm Initialization and Execution:

- Within the Segment Editor, select the Grow from Seeds effect.
- Click Initialize from current seeds to configure the algorithm based on the annotated seed regions.
- Execute the segmentation by clicking Apply. The algorithm employs a probabilistic region-growing approach, propagating segment labels from seed points while respecting intensity gradients and spatial continuity constraints.
- Processing time may vary depending on dataset size and computational resources.

Upon completion, inspect the resulting segmentation. If overlapping segments or incomplete delineations are observed, refine the seed distribution by adding additional markers to underrepresented regions and re-run the algorithm.

5. Three-Dimensional Visualization and Quality Assessment

- Navigate to the Segmentations module. Enable Segmentation preview and configure the display mode as Closed surface for three-dimensional rendering. Use the Recenter view function to optimize the visualization perspective.
- Temporarily hide or remove the auxiliary "Other" segment to isolate and evaluate the primary cardiac structure

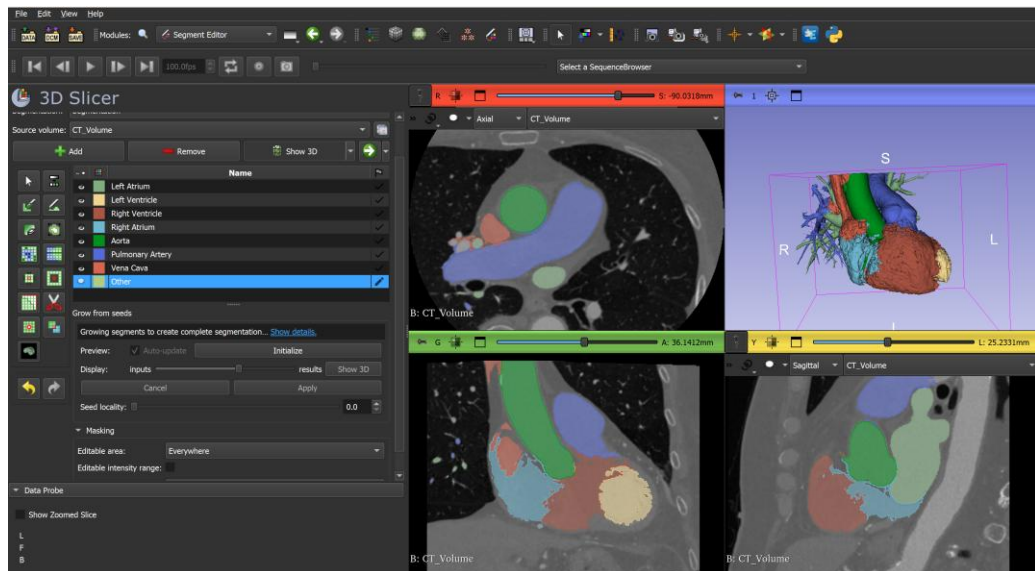


Figure 4. Resulting whole heart segmentation after the process of automated segmentation using "Grow from seeds" instrument.

6. Refinement of Left Atrial Segmentation.

6.1. Isolation and Manual Correction

- Hide all segments except Left Atrium. The initial segmentation may exhibit surface irregularities, spurious vascular extensions, or topological artifacts. Utilize the Erase tool with a spherical brush in the 3D view to manually remove non-physiological protrusions while preserving anatomical fidelity

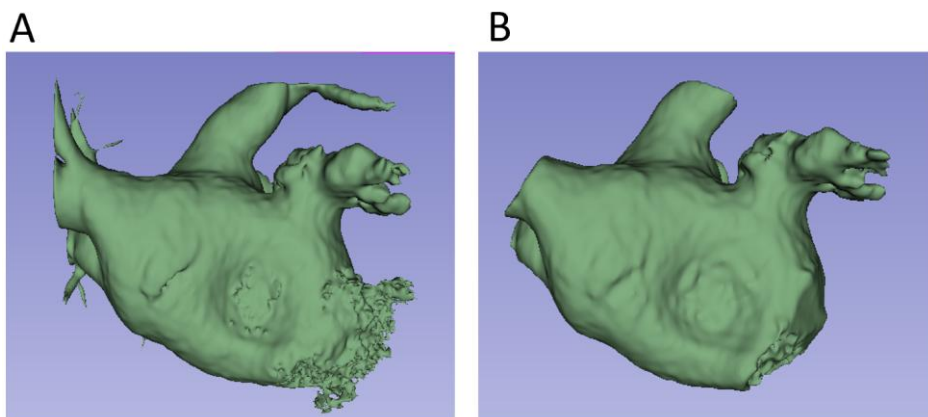


Figure 5. Left atrium segmentation (A) before surface regularization; (B) after removing vascular extensions and surface smoothing.

6.2. Endocardial Surface Extraction via Thresholding

- Apply a Threshold effect to the Left Atrium segment to generate a binary mask corresponding to the blood pool and endocardial boundary. Adjust threshold values based on Hounsfield unit distributions characteristic of contrast-enhanced blood.
- Generate a new empty segment named Left Atrium initial. Use the Logical operators effect to copy the thresholded endocardial mask into this segment.

6.3. Wall Thickness Reconstruction via Morphological Operations

- Select the original Left Atrium segment. Apply the Margin effect with the Grow operation, using a step size equal to the isotropic voxel spacing (e.g., 0.4 mm). Iteratively apply dilation until the segmented boundary stabilizes, indicating convergence with the epicardial surface.

- Perform a Boolean subtraction: Left Atrium (dilated) – Left Atrium initial = Left Atrium wall. Remove the temporary threshold mask
- Apply Smoothing → Joint smoothing with a smoothing factor of 0.35 to eliminate sharp artifacts and ensure geometric continuity. Use the Islands effect with Remove small islands to eliminate disconnected voxel clusters arising from segmentation noise.

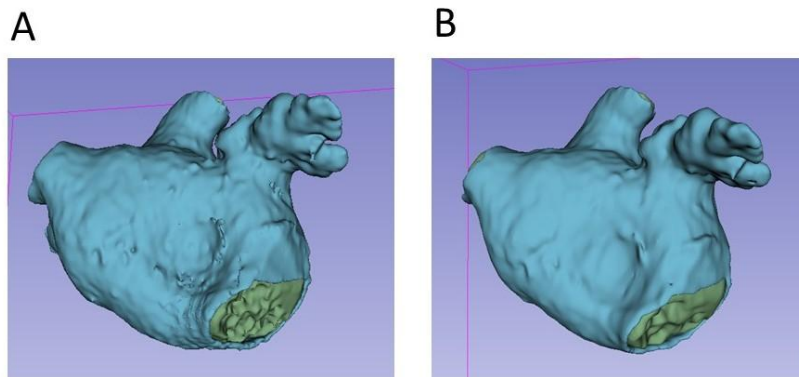


Figure 6. Blood pool and atrial wall segmentation: (A) before topological cleaning; (B) after topological cleaning.

6.4. Epicardial surface assembly:

- Create a new empty segment. Use Logical operators → Add to combine Left Atrium wall and Left Atrium initial, yielding the final epicardial surface representation.

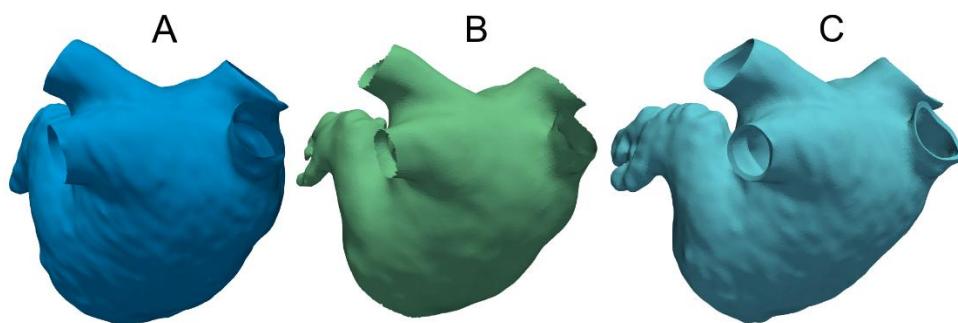


Figure 7. Resulting atrial models: (A) Epicardial layer; (B) Endocardial layer; (C) Atrial wall.

7. Final Output Generation

- The protocol yields three distinct geometric representations:
 1. Endocardial surface: Blood–tissue interface of the left atrium.
 2. Epicardial surface: Outer myocardial boundary.
 3. Myocardial wall: Volumetric representation of atrial tissue thickness.
- Export each segment as a standalone 3D model:
 1. Right-click the segmentation node → Export to files.
 2. Select Model format with .vtk extension (uncompressed ASCII or binary as required).
 3. Ensure coordinate system consistency and metadata preservation for downstream electrophysiological modeling.

Applications and Downstream Processing

The resulting patient-specific geometric models are suitable for:

- Anisotropic fiber orientation mapping and conduction velocity modeling
- Computational electrophysiology simulations (e.g., action potential propagation, rotor dynamics)
- Morphometric analysis of atrial remodeling in atrial fibrillation

Note: All segmentation steps should be documented and version-controlled to ensure reproducibility. Validation against expert manual annotations or high-resolution ex vivo imaging is recommended for quantitative assessment of segmentation accuracy.